Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (EVEN)**	**Course Name**	**AIML Lab**
Semester	**6**	**Course Code**	**23CT1602**
Roll No	**23**	**Name of Student**	**Srushti Chaudhari**

Practical Number	1-B
Course Outcome	CO1: Apply Artificial Intelligence problem formulation techniques, intelligent agents, search algorithms, and logical inference to solve real-world problems. CO2: Implement and evaluate machine learning algorithms including classification, clustering, and dimensionality reduction using appropriate preprocessing and performance metrics. CO3: Analyze and compare ensemble learning models.
Aim	Two-Room Vacuum Cleaner Agent Simulation To design and implement a **Vacuum Cleaner Agent** that operates in a **two-room environment (Room A and Room B)** and cleans the rooms using simple condition-based rules.
Problem Definition	A vacuum cleaner agent operates in an environment consisting of two rooms: **Room A and Room B**. Each room may either be **Clean** or **Dirty**. The agent starts in **Room A** and follows simple rules to clean the environment. If the current room is dirty, the agent cleans it. If the room is already clean, the agent moves to the other room. The agent continues this process until **both rooms are clean**. The goal is to simulate how an intelligent agent interacts with its environment and performs actions based on percepts to achieve the desired objective.
Theory (100 words)	A **Vacuum Cleaner Agent** is a classic example used in Artificial Intelligence to demonstrate how agents interact with environments. The agent perceives the state of the environment through sensors and performs actions using actuators. In a **two-room environment**, the vacuum agent checks whether the current room is clean or dirty. If the room is dirty, it performs the **suck action** to clean it. If the room is clean, it moves to the other room. The agent continues this process until both rooms are clean. This model helps illustrate the concept of **state, percept, action, and environment** in intelligent agent design.

Procedure and Execution

(100 Words)

Algorithm:

1. Start the program.
2. Initialize the environment with two rooms: **Room A and Room B**.
3. Set the initial location of the agent in **Room A**.
4. Check the status of the current room.
5. If the room is **Dirty**, clean the room.
6. If the room is **Clean**, move to the other room.
7. Repeat the process until **both rooms are clean**.
8. Display the actions performed by the agent.
9. Stop the program.

Code:

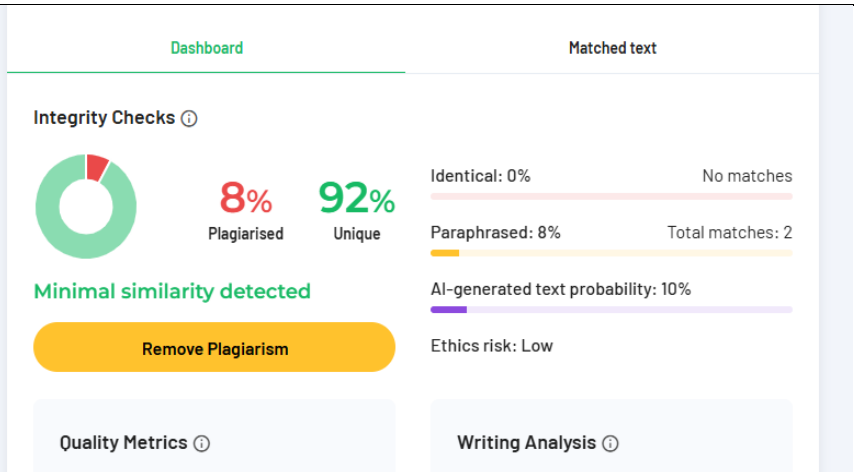
```
rooms = {"A": "Dirty", "B": "Dirty"}
current_room = "A"

while True:
print("Current Room:", current_room)
print("Room Status:", rooms)

if rooms[current_room] == "Dirty":
print("Action: Cleaning Room", current_room)
rooms[current_room] = "Clean"
else:
print("Room is Clean")

if rooms["A"] == "Clean" and rooms["B"] == "Clean":
print("Both rooms are clean. Stopping agent.")
break

if current_room == "A":
current_room = "B"
else:
current_room = "A"
```

Output:

```
•••  Current Room: A
     Room Status: {'A': 'Dirty', 'B': 'Dirty'}
     Action: Cleaning Room A
     Current Room: B
     Room Status: {'A': 'Clean', 'B': 'Dirty'}
     Action: Cleaning Room B
     Both rooms are clean. Stopping agent.
```

Output Analysis

The vacuum cleaner agent successfully cleans both rooms by checking the condition of the current room and performing the appropriate action. If the room is dirty, it cleans it; otherwise, it moves to the other room. This demonstrates how a simple intelligent agent interacts with its environment and performs actions to achieve its goal.

Link of student Github profile where lab assignment has	https://github.com/SRchaudhari15/AIML-Practicals.git
Conclusion	The Two-Room Vacuum Cleaner Agent simulation was successfully implemented. The agent followed simple condition-action rules to clean both rooms efficiently. This experiment demonstrates how intelligent agents perceive their environment and perform actions to achieve a goal.
Plag Report (Similarity index < 12%)	
Date	29/12/25

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